```
0ZHNmVNMVAxUU09IiwiYWxnIjoiUlMyNTYifQ.eyJhdF9oYXNoIjoiTXF2SDY1
NngyU0wzc2dnT25yZmNkZyIsInN1YiI6InRlc3QiLCJhdWRpdFRyYWNraW5nSW
QiOiIxNDViNjI2MC1lNzA2LTRkNDctYWVmYy1lMDIzMTQyZjBjNjMtMzg2MTki
LCJpc3MiOiJodHRwOi8vYXMuZXhhbXBsZS5jb206ODA4MC9vYXV0aDIiLCJ0b2
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YXV0aF90aW1lIjoxNTgxMzc1MzI1LCJyZWFsbSI6Ii8iLCJleHAiOjE1ODEzNz
g5MjYsInRva2VuVHlwZSI6IkpXVFRva2VuIiwiaWF0IjoxNTgxMzc1MzI2fQ.S
5Ib5Acj5hZ7se9KvtlF2vpByG_0XAWKSg0-
Zy_GZmpatrox0460u5HYvPdOVl7qqP-
AtTV1ah_2aFzX1qN99ituo8fOBIpKDTyEgHZcxeZQDskss1QO8ZjdoE-
JwHmzFzIXMU-5u9ndfX7--
Wu_QiuzB45_NsMi72ps9EP8iOMGVAQyjFG5U6jO7jEWHUKI87wrv1iLjaFUcG0
H8YhUIIPymk-CJUgwtCBzESQ1R7Sf-6mpVgAjHA-eQXGjH18tw1dRneq-kY-
D1KU0wxMnw0GwBDK-
LudtCBaETiH5T_CguDyRJJotAq65_MNCh0mhsw4VgsvAX5Rx30FQijXjNw","t
oken_type":"Bearer","expires_in":3599}
```


A.3.2 Installing an open source AS

To be written.

A.4 Installing an LDAP directory server

An LDAP directory server is needed for some of the examples in chapter 8. As for the OAuth2 AS, I've provided instructions for a commercial and open source LDAP server.

> **TIP** Apache Directory Studio is a useful tool for browsing LDAP directories. It can be downloaded from https://directory.apache.org/studio/.

A.4.1 ForgeRock Directory Services